

Wonjun Lee

PH.D. CANDIDATE IN ELECTRICAL ENGINEERING, CILAB, KAIST

291, Daehak-ro, Yuseong-gu, Daejeon, 34141, Rep. of KOREA

☎ (+82) 10-5420-3553 | ✉ dpenguin2000[at]gmail.com | 📧 Wonjun-Lee1009 | 🎓 Google Scholar

I am currently pursuing a Ph.D. degree in Electrical Engineering at KAIST. My academic journey began at Gyeonggi Science High School for the Gifted, which fostered my passion for research and innovation. I then earned my B.S. in Electrical Engineering and Computer Science at KAIST, and M.S. in Electrical Engineering at KAIST. I enjoy engaging with creative challenges and value building meaningful relationships along the way. Please feel free to explore my academic background below and contact me directly for further details.

Research Interest

As AI's role expands across various fields, building **trustworthy AI** is crucial. I am particularly interested in AI security, as vulnerabilities compromise reliability and fairness.

Additionally, as humans rely on various senses for judgment, I focus on research in **Multi-Modal Learning** and **AI Security**. My goal is to help bridge the understanding between AI and humans. My work explores the following, but not limited to:

- **Multi-Modal Learning**: multi-modal large language models, efficient multi-modal learning
- **AI Security**: adversarial attack, jailbreaking and their defense
- **Multimedia Forensics**: detecting forgeries, synthetic images, and deepfakes

Education

Korea Advanced Institute of Science and Technology (KAIST)

PH.D. IN ELECTRICAL ENGINEERING

Daejeon, S.Korea

Mar. 2025 - Present

Korea Advanced Institute of Science and Technology (KAIST)

M.S. IN ELECTRICAL ENGINEERING

Daejeon, S.Korea

Mar. 2023 - Feb. 2025

Korea Advanced Institute of Science and Technology (KAIST)

B.S. IN ELECTRICAL ENGINEERING & SCHOOL OF COMPUTING (DOUBLE MAJOR)

Daejeon, S.Korea

Mar. 2019 - Feb. 2023

Gyeonggi Science High School for the Gifted (GSHS)

MAJOR IN MATH, PHYSICS, AND BIOLOGY

Suwon, Gyeonggi-do, S.Korea

Mar. 2016 - Feb. 2019

Publication

INTERNATIONAL

AIM: Anchor Identity Features, then Match for Multimodal Large Language Model Unlearning

2026

THE 2026 CONFERENCE ON EMPIRICAL METHODS IN NATURAL LANGUAGE PROCESSING (EMNLP FINDINGS)

Multimodal LLM Unlearning

Wonjun Lee*, Jaehyuk Jang*, Kangwook Ko*, Hee-Seon Kim, Changick Kim (*: Equal Contribution)

Where Identity Lives: Localized, Retain-Free Identity Unlearning in Multimodal Large Language Models

2026

THE 2026 CONFERENCE ON EMPIRICAL METHODS IN NATURAL LANGUAGE PROCESSING (EMNLP FINDINGS)

Multimodal LLM Unlearning

Kangwook Ko*, Jaehyuk Jang*, Wonjun Lee*, Hee-Seon Kim, Changick Kim (*: Equal Contribution)

Constraining to Generalize: Subspace Tuning for Few-shot Generalization of Audio-Language Models

2026

THE 2026 CONFERENCE ON EMPIRICAL METHODS IN NATURAL LANGUAGE PROCESSING (EMNLP FINDINGS)

ALM Prompt Tuning

Jaehyuk Jang, Kangwook Ko, Wonjun Lee, Changick Kim

Safety in Batches? Understanding and Mitigating Safety Failures in Batch Prompting

2026

PREPRINT

LLM Safety

Kihyun Kim, Hee-Seon Kim, Wonjun Lee, Changick Kim

Jailbreak to Protect: Buffering Harmful Fine-Tuning via Temporary Jailbreaking LoRA in Large Language Models

2026

INTERNATIONAL CONFERENCE ON MACHINE LEARNING (ICML) **SPOTLIGHT (TOP 2.2%)**

Multimodal LLM Safety

Seokil Ham, Jaehyuk Jang, Wonjun Lee, Changick Kim

Generalizable Prompt Tuning for Audio-Language Models via Semantic Expansion

2026

THE 64TH ANNUAL MEETING OF THE ASSOCIATION FOR COMPUTATIONAL LINGUISTICS (ACL FINDINGS)

ALM Prompt Tuning

Jaehyuk Jang*, Wonjun Lee*, Kangwook Ko*, Changick Kim (*: Equal Contribution)

Efficient Test-Time Optimization for Depth Completion via Low-Rank Decoder Adaptation

PREPRINT

Minseok Seo*, **Wonjun Lee***, Jaehyuk Jang, Changick Kim (*: Equal Contribution)

2026
Depth Estimation, VFM

SELF: Selective Fusion of Identity for Generalizable Deepfake Detection

PREPRINT

Younghun Kim, Minsuk Jang, Myung-Joon Kwon, **Wonjun Lee**, Changick Kim

2025
Deepfake Detection

Benign-to-Toxic Jailbreaking: Inducing Harmful Responses from Harmless Prompts

PREPRINT

Hee-Seon Kim, Minbeom Kim, **Wonjun Lee**, Kihyun Kim, Changick Kim

2025
Multimodal LLM Safety

SAFIRE: Segment Any Forged Image Region

AAAI CONFERENCE ON ARTIFICIAL INTELLIGENCE (AAAI)

Myung-Joon Kwon*, **Wonjun Lee***, Seung-Hun Nam, Minji Son, Changick Kim (*: Equal Contribution)

2025
Image Forgery Localization

FRIDAY: Mitigating Unintentional Facial Identity in Deepfake Detectors Guided by Facial Recognizers

VISUAL COMMUNICATIONS AND IMAGE PROCESSING (VCIP) **ORAL PRESENTATION**

Younghun Kim, Myung-Joon Kwon, **Wonjun Lee**, Changick Kim

2024
Deepfake Detection

DOMESTIC

Image-Text Order Matters: Input Ordering Effects in LVLM Jailbreaks

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE) **LG ELECTRONICS PAPER AWARD**

Kihyun Seol, **Wonjun Lee**, Kihyun Kim, Minbeom Kim, Changick Kim

2026
Multimodal LLM Safety

Safety Analysis of Image Verification-Based Reasoning Methods in MLLMs

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE)

Jihyun Lee, **Wonjun Lee**, Kihyun Kim, Hee-Seon Kim, Changick Kim

2026
Multimodal LLM Safety

Image-Based Universal Jailbreak Attack via Safety Boundary Approximation

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE)

Kihyun Kim, Hee-Seon Kim, Minbeom Kim, **Wonjun Lee**, Changick Kim

2025
Multimodal LLM Safety

Ensembling Pretrained Models for Enhanced Generalization in Synthetic Image Detection

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE) **LG ELECTRONICS PAPER AWARD**

Myung-Joon Kwon, **Wonjun Lee**, Younghun Kim, Changick Kim

Synthetic Image Detection

Combating Bias: Corrective Measures Through Adversarial Training in Neural Networks

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE) **LG ELECTRONICS PAPER AWARD**

Wonjun Lee, Myung-Joon Kwon, Younghun Kim, Changick Kim

2024
De-Biasing, Adversarial Training

Improving the Generalization Performance of Deepfake Detection Through a Partial Image Manipulation Augmentation Technique

INSTITUTE OF ELECTRONICS AND INFORMATION ENGINEERS (IEIE)

Younghun Kim, Myung-Joon Kwon, **Wonjun Lee**, Changick Kim

2024
Deepfake Detection

Research Experience

Roen Surgical

KAIST EE EXTERNSHIP PROGRAM SECOND COHORT

- Participated as a key developer in the patent development of "Stone size estimation method."
- Contributed to technical development and validation for image-based stone size estimation in surgical applications.

Daejeon, S.Korea
Jun.2022 – Dec.2022

SAMSUNG Electronics CE/IM Division Mobile Communications Unit

SUMMER INTERNSHIP

- Worked on Galaxy Tab FE design.
- Gained hands-on experience with mobile wireless communication module design and product development workflows.

Remote
Jul.2021 - Aug.2021

SAMSUNG Electronics DS Division Foundry Business Unit

SAMSUNG TALENT INTERNSHIP PROGRAM (STIP)

- Worked on DRAM design.
- Gained hands-on experience with semiconductor memory design flows and foundry development workflows.

Hwaseong, Gyeonggi-do, S.Korea
Jul.2020 - Aug.2020

Patent

A Universal Image-Generation Framework for Jailbreaking Large Vision-Language Models by Bypassing Safety Alignment	2025
KR PATENT 10-2025-0157353	
- Participated as a Key Developer in Patent Development - Participated through the Project with ETRI	
Method and System for Generating Universal Adversarial Perturbations Using High-Sensitivity Components of Vision Encoders in Large-Scale Vision-Language Models	2025
KR PATENT 10-2025-0194468	
- Participated as a Key Developer in Patent Development - Participated through the Project with IITP	
Stone size estimation method	2023
KR PATENT 10-2023-0031846	
- Participated as a Key Developer in Patent Development - Participated through the KAIST EE Externship	

Honors & Awards

2026	LG Electronics Paper Award , Summer Annual Conference of IEIE	Jeju, S.Korea
2024	LG Electronics Paper Award (2 papers) , Summer Annual Conference of IEIE	Jeju, S.Korea
2023	Certificate , KAIST Leadership and Volunteer Excellence Award Certificate	Daejeon, S.Korea
2018	Bronze Award , SAMSUNG Humantech Paper Award Biology Division	Seoul, S.Korea

Scholarship

Korea National Scholarship (B.S.)	Mar. 2019 - Feb. 2023
Korea National Scholarship (M.S.)	Mar. 2023 - Feb. 2025
Korea National Scholarship (Ph.D.)	Mar. 2025 - Present

Project

Automatic Evaluation Benchmark Generation from LLM Log Data	Jun. 2025 - Aug. 2030
SAMSUNG ELECTRONICS DS	
Development of AI Technology with Robust and Flexible Resilience Against Risk Factors	Jan. 2025 - Dec. 2028
ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE (ETRI)	
Team Leader	
Penetration Security Testing of ML Model Vulnerabilities and Defense	Jan. 2025 - Dec. 2027
INSTITUTE FOR INFORMATION & COMMUNICATION TECHNOLOGY PLANNING & EVALUATION (IITP)	
Scene Text Recognition with Visual Contexts	Sep. 2024 - Dec. 2024
CENTER FOR SECURITY TECHNOLOGY RESEARCH, KAIST	
Bypass Techniques for Identifying Vulnerabilities in CAPTCHA	Mar. 2023 - Dec. 2023
CENTER FOR SECURITY TECHNOLOGY RESEARCH, KAIST	
Practical Adversarial Attacks of AI Facial Recognition Technology Using Physical Patterns	Mar. 2023 - Dec. 2023
CENTER FOR SECURITY TECHNOLOGY RESEARCH, KAIST	

Academic Service

Visual Communications and Image Processing (VCIP)	2025 - Present
AAAI Conference on Artificial Intelligence (AAAI)	2026 - Present
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)	2026 - Present

KAIST Electrical Engineering TA MEMBER • Signals and Systems - Fall 2023 • Introduction to Electronics Design Lab - Fall 2023, Fall 2024, Spring 2025, Spring 2026 • Digital Image Processing - Spring 2024 • EE Career Development I - Spring 2025, Spring 2026 • Introduction to Machine Learning - Fall 2025	<i>Daejeon, S.Korea</i> <i>Sep. 2023 - Present</i>
KAIST Student Counseling Assistant (CA) MEMBER • The CA (Counseling Assistant) program aims to expand counseling support opportunities for students and graduate students to ensure a smooth academic/school life and career planning. • Participants work autonomously with peers to enhance their development skills and foster personal growth.	<i>Daejeon, S.Korea</i> <i>Mar. 2024 - Feb. 2025, Sep. 2025 - Present</i>
CILAB, KAIST SERVER VICE TEAM LEADER VICE LAB LEADER SERVER TEAM LEADER	<i>Daejeon, S.Korea</i> <i>Nov. 2025 - Nov. 2026</i> <i>Nov. 2025 - Nov. 2026</i> <i>Nov. 2024 - Nov. 2025</i>
KAIST Electrical Engineering Student Council VICE DEPARTMENT REPRESENTATIVE	<i>Daejeon, S.Korea</i> <i>Feb. 2020 - Feb.2022</i>
KAIST Dance Crew Illusion PRESIDENT	<i>Daejeon, S.Korea</i> <i>Jan. 2020 - Feb.2022</i>
MadCamp 11th Cohort Hosted by KAIST CS MEMBER • MadCamp is an intensive program in collaboration with KAIST, focused on app, web, and game development for undergraduate students in South Korea. • Participants work autonomously with peers to enhance their development skills and foster personal growth.	<i>Daejeon, S.Korea</i> <i>Dec. 2020 - Jan. 2021</i>